Foreword

Although it is still fairly young, the software development industry has matured considerably in the last 15 years and gone through several major transformations:

- Just a short while ago, it seems, the waterfall lifecycle was pretty much the only option for software development projects. Today, agile methodology is also frequently used.
- New development engineering practices have entered the game, such as SCM, issue tracking, build standardization, continuous integration, continuous inspection, and so on. In most organizations, it is now the norm to have a complete software factory.
- Although they started out minimalistic, modern IDEs have become a widely adopted productivity tool for developers.

This is all great news; and what's more, there is strong traction to continue these efforts and make the software development world even better. It is amazing how many development teams are seeking a common Holy Grail: continuous delivery. In other words, teams want a software development process that is predictable and repeatable, and that enables a shift to production at any time in a controlled manner.

Despite all the good things that have happened within the industry in recent years, there is a general consensus that we are not quite there yet. Software development is not yet a fully mastered science, and delivery generally still is not reliable. Projects are often delivered late, with a reduced scope of features, generating frustration at all levels in organizations and justifying their reputation for being both expensive and unpredictable.

One aspect that is missing from the recent improvements in our industry is *measurement*: measurement of what we produce, of course, but also measurement of the impact of the changes we make to improve delivery. We should be able to answer questions such as, "Did this change improve the process?" and "Are we delivering better now?" In many cases, these